

STEFAN CUTLER

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EDUCATION

Olin College of Engineering | Needham, MA

Electrical and Computer Engineering | First-year student

National Merit Commended Student, 2026

EXPERIENCE

Synk (synk.today)

Software Engineering & Machine Learning Intern

- Improved efficiency of C++ and Python automation code for Synk, an AI and virtual-experiences platform for browser-based networking events.

Olin College of Engineering | 2026–Present

AI Research with Dr. Elisabeth Sylvan

- Investigated whether targeted questions and AI-written instructions for a second AI improve planning efficiency compared with built-in AI plan mode.

Olin Electric Motorsports | 2026–Present

AIR Control Firmware | High Voltage Subteam

- Ported AIR (Accumulator Isolation Relay) control firmware from ATmega16M1 to STM32G441 in C, covering precharge, contactor feedback, shutdown checks, CAN monitoring, and fault handling. The local branch compiles for ARM and passes host tests; board pin mapping and hardware validation pass as well.

FIRST Robotics Competition Team 2638 | Late 2024–Early 2026

Build Team Member

- Contributed to competition robot construction; welding, cutting, cad pieces; team competed at the FIRST Championship in both seasons.

PROJECTS

Smart Mirror | 2026

Raspberry Pi 4 (4 GB), MagicMirror²

- Built a working smart mirror around a Raspberry Pi 4 (4 GB) and repurposed monitor, adapting the MagicMirror² open-source display.
- Added a Canvas section that shows upcoming assignments and due dates directly on the mirror.

Stackwise Tolerance Calculator | Early 2026–Present

React, TypeScript

- Led development of a React/TypeScript tolerance tool with signed dimension stacks, inch/mm conversion, worst-case limits, RSS estimates, and component-contribution ranking.
- Built seeded Monte Carlo analysis with normal, uniform, and triangular inputs; the app reports yield and sensitivity and supports CSV/JSON project exchange.

DECK — DECA Practice Application | Early 2026–Present

UI and Data Storage Contributor

- Centralized 21 instructional-area labels in shared JSON and added sync checks to flag missing labels between the interface and question bank.
- Extended Python bank verification to catch missing files and manifest-count mismatches; fixed filtered reports to recompute summary counters.

Temperature-Sensing Prototype | Early 2026–Present

Arduino Uno

- Read a four-pin sensor's AO signal on Arduino Uno; used serial output to diagnose readings that fell as the sensor warmed while prototyping relay and LED switching.

SKILLS AND ACTIVITIES

Programming: C, C++, TypeScript/JavaScript; Python data-validation tooling

Embedded: STM32G441, Arduino Uno, GPIO, FDCAN, sensor and serial debugging

Tools and activities: React, Next.js, Git, GitHub, VS Code; ICPC participation and C++ algorithm practice